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## Integrated Circuit Device And Oscillator

### Abstract

An integrated circuit device includes a clock input terminal disposed along a first side, to which a reference clock signal is input, a first power supply voltage input terminal disposed along the first side, to which a first power supply voltage is input, a clock output terminal disposed along the second side, from which an output clock signal having a frequency different from that of the reference clock signal is output, a second power supply voltage input terminal disposed along the second side, to which a second power supply voltage is input, a first ground terminal disposed between the clock input terminal and the first power supply voltage input terminal along the first side, and a second ground terminal disposed between the clock output terminal and the second power supply voltage input terminal along the second side.

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## Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-028240, filed Feb. 28, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND
1. Technical Field
[0002] The present disclosure relates to an integrated circuit device and an oscillator.
2. Related Art
[0003] JP-A-2012-124630 discloses an integrated circuit in which power supply pads that supply a power supply voltage to an integer division circuit, a decimal division circuit, a phase comparator, and a charge pump are separately provided in a fractional frequency division PLL circuit, and thereby, coupling of frequency components that cause spurious via a power supply line is prevented.
[0004] JP-A-2012-124630 is an example of the related art.
[0005] In the integrated circuit disclosed in JP-A-2012-124630, interference noise propagating in the PLL circuit can be suppressed, however, an effect of suppressing interferences via terminals is not expected.
SUMMARY
[0006] An integrated circuit device according to an aspect of the present disclosure having a first side and a second side as an opposite side to the first side in a plan view, includes a clock input terminal disposed along the first side, to which a reference clock signal is input, a first power supply voltage input terminal disposed along the first side, to which a first power supply voltage as a source of a power supply voltage supplied to a first circuit that operates at a frequency of the reference clock signal is input, a clock output terminal disposed along the second side, from which an output clock signal having a frequency different from that of the reference clock signal is output, a second power supply voltage input terminal disposed along the second side, to which a second power supply voltage as a source of a power supply voltage supplied to a second circuit that operates at the frequency of the output clock signal is input, a first ground terminal disposed between the clock input terminal and the first power supply voltage input terminal along the first side, and a second ground terminal disposed between the clock output terminal and the second power supply voltage input terminal along the second side.
[0007] An oscillator according to an aspect of the present disclosure includes the integrated circuit device according to the aspect.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a cross-sectional view of an oscillator of an embodiment.
[0009] FIG. **2** is a plan view of the oscillator of the embodiment.
[0010] FIG. **3** is a cross-sectional view showing an inner package of the oscillator and the inside thereof.
[0011] FIG. **4** is a cross-sectional view showing a resonator of the oscillator.
[0012] FIG. **5** is a functional block diagram of an oscillator of a first embodiment.
[0013] FIG. **6** shows a configuration example of a fractional N-PLL circuit.
[0014] FIG. **7** shows a configuration example of a PLL circuit.
[0015] FIG. **8** shows a layout configuration of a control IC in the first embodiment.
[0016] FIG. **9** shows electromagnetic field coupling.

[0017] FIG. **10** shows capacitive coupling in the first embodiment.
[0018] FIG. **11** shows a layout configuration of a control IC in a second embodiment.
[0019] FIG. **12** shows capacitive coupling in the second embodiment.

DESCRIPTION OF EMBODIMENTS

[0020] As below, preferred embodiments of the present disclosure will be described in detail using the drawings. Note that the embodiments to be described below do not unduly limit the present disclosure described in What is Claimed is. In addition, not all configurations to be described below are necessarily essential component elements of the present disclosure.

1. First Embodiment

1-1. Configuration of Oscillator

[0021] FIG. **1** is a cross-sectional view showing an oscillator of an embodiment. FIG. **2** is a plan view of the oscillator as seen from above. FIG. **3** is a cross-sectional view showing an inner package of the oscillator and the inside thereof. FIG. **4** is a cross-sectional view showing a resonator of the oscillator.

[0022] An oscillator **1** shown in FIGS. **1** and **2** is an oven-controlled crystal oscillator, and includes an outer package **2**, an inner package **3**, a control IC **4**, and a resonator **5**. The inner package **3**, the control IC **4**, and the resonator **5** are housed in the outer package **2**.

[0023] As shown in FIG. **1**, the outer package **2** includes an outer base **21** and an outer lid **22**. The outer base **21** includes a board **27**, a frame-shaped wall portion **28** standing upward from an edge portion of an upper surface of the board **27**, and a frame-shaped leg portion **29** standing downward from an edge portion of a lower surface of the board **27**. The upper surface of the board **27** and the wall portion **28** form an upper recess **211** that opens in an upper surface **21***a* of the outer base **21**, and the lower surface of the board **27** and the leg portion **29** form a lower recess **212** that opens in a lower surface **21***b* of the outer base **21**. Therefore, the outer base **21** has a substantially H-shaped cross section.

[0024] The upper recess **211** includes a first upper recess **211***a* that opens in the upper surface **21***a*, a second upper recess **211***b* that opens in a bottom surface of the first upper recess **211***a* and has an opening smaller than that of the first upper recess **211***a*, and a third upper recess **211***c* that opens in a bottom surface of the second upper recess **211***b* and has an opening smaller than that of the second upper recess **211***b*. The control IC **4** is disposed at the bottom surface of the first upper recess **211***a*, and the inner package **3** is disposed at a bottom surface of the third upper recess **211***c*.

[0025] The outer lid **22** is bonded to the upper surface **21***a* of the outer base **21** via a sealing member **23** such as a seal ring or low-melting-point glass to close the opening of the upper recess **211**. Accordingly, the upper recess **211** is hermetically sealed, and an outer housing space S**2** as a housing space is formed in the outer package **2**. Meanwhile, the opening of the lower recess **212** is not sealed and faces the outside of the outer package **2**. The inner package **3** and the control IC **4** are housed in the outer housing space S**2**, and the resonator **5** is disposed at the lower recess **212**.

[0026] The outer base **21** includes a plurality of internal terminals **241** disposed at the bottom surface of the first upper recess **211***a*, a plurality of internal terminals **242** disposed at the bottom surface of the second upper recess **211***b*, a plurality of internal terminals **243** disposed at a bottom surface of the lower recess **212**, and a plurality of external terminals **244** disposed at the lower surface **21***b*, that is, a top surface of the leg portion **29**. Each internal terminal **241** is electrically coupled to the control IC **4** via a bonding wire BW**1**, each internal terminal **242** is electrically coupled to the inner package **3** via a bonding wire BW**2**, and each internal terminal **243** is electrically coupled to the resonator **5** via a conductive bonding member B**1**.

[0027] The respective terminals **241**, **242**, **243**, and **244** are electrically coupled as appropriate through internal wiring **25** formed in the outer base **21** to electrically couple the control IC **4**, the inner package **3**, the resonator **5**, and the external terminals **244**. The internal wiring **25** is coupled to the external terminals **244** through the inside of the leg portion **29**. In the external terminals **244**, coupling to an external apparatus (not shown) is made. A side surface terminal **245** coupled to the external terminal **244** is disposed at a side surface of the leg portion **29**. The side surface terminal **245** is a castellation. Therefore, solder H spreads over the side surface terminal **245** to form a fillet, and thus mechanical and electrical bonding with the external apparatus becomes stronger. However, the configuration is not limited thereto, and for example, the side surface terminal **245** may be omitted.

[0028] As shown in FIG. **3**, the inner package **3** includes an inner base **31** and an inner lid **32**. The inner base **31** has a recess **311** that opens in a lower surface **31***b*.

[0029] The recess **311** includes a first recess **311***a* that opens in the lower surface **31***b*, a second recess **311***b* that opens in a bottom surface of the first recess **311***a* and has an opening smaller than that of the first recess **311***a*, and a third recess **311***c* that opens in a bottom surface of the second recess **311***b* and has an opening smaller than that of the second recess **311***b*. A resonator element **6** is disposed at the bottom surface of the first recess **311***a*, and a heat generation IC **7** and an oscillation IC **8** are disposed side by side in an X-axis direction at a bottom surface of the third recess **311***c*.

[0030] The inner lid **32** is bonded to the lower surface **31***b* of the inner base **31** via a sealing member **33** such as a seal ring or low-melting-point glass to close the opening of the recess **311**. Accordingly, the recess **311** is hermetically sealed, and an inner housing space S**3** is formed in the inner package **3**. The resonator element **6**, the heat generation IC **7**, and the oscillation IC **8** are housed in the inner housing space S**3**.

[0031] The inner housing space S**3** is airtight and in a reduced pressure state, and is preferably in a state closer to vacuum. Accordingly, viscous resistance in the inner housing space S**3** is reduced, and a vibrational characteristic of the resonator element **6** is improved. However, the atmosphere in the inner housing space S**3** is not particularly limited.

[0032] The inner base **31** includes a plurality of internal terminals **341** disposed at the bottom surface of the first recess **311***a*, a plurality of internal terminals **342** and **343** disposed at the bottom surface of the second recess **311***b*, and a plurality of external terminals **344** disposed at an upper surface **31***a* of the inner base **31**. Each internal terminal **341** is electrically coupled to the resonator element **6** via a conductive bonding member B**2** and a bonding wire BW**3**, each internal terminal **342** is electrically coupled to the heat generation IC **7** via a bonding wire BW**4**, and each internal terminal **343** is electrically coupled to the oscillation IC **8** via a bonding wire BW**5**.

[0033] The terminals **341**, **342**, **343**, and **344** are electrically coupled as appropriate through internal wiring (not shown) formed in the inner package **3** to electrically couple the resonator element **6**, the heat generation IC **7**, the oscillation IC **8**, and the external terminal **344**. The inside and the outside of such an inner package **3** are electrically coupled via the external terminal **344**.

[0034] The inner package **3** as described above is fixed to the bottom surface of the third upper recess **211***c* via a bonding member B**3** having sufficiently low thermal conductivity at the inner lid **32**.

[0035] As shown in FIG. **3**, the heat generation IC **7** is disposed at the bottom surface of the third recess **311***c* with an active surface facing downward (toward the inner lid **32**) and is electrically coupled to the plurality of internal terminals **342** via the bonding wire BW**4**. The oscillation IC **8** is disposed at the bottom surface of the third recess **311***c* with an active surface facing downward (toward the inner lid **32**) and is electrically coupled to the plurality of internal terminals **343** via the bonding wire BW**5**.

[0036] As shown in FIG. **4**, the resonator **5** includes a package **51** and a resonator element **55** housed in the package **51**.

[0037] The package **51** includes a base **52** and a lid **53**. The base **52** has a recess **521** that opens in a lower surface **52***b*. The resonator element **55** is disposed at a bottom surface of the recess **521**.

[0038] The lid **53** is bonded to the lower surface **52***b* of the base **52** via a sealing member **54** such as a seal ring or low-melting-point glass to close the opening of the recess **521**. Accordingly, the recess **521** is hermetically sealed, and a housing space S**5** is formed in the package **51**. The resonator element **55** is housed in the housing space S**5**. The housing space S**5** is airtight and in a reduced pressure state, and is preferably in a state closer to vacuum. Accordingly, viscous resistance in the housing space S**5** is reduced, and a vibrational characteristic of the resonator element **55** is improved. However, the atmosphere in the housing space S**5** is not particularly limited.

[0039] The base **52** includes a plurality of internal terminals **561** disposed at the bottom surface of the recess **521** and a plurality of external terminals **564** disposed at an upper surface **52**a of the base **52**. Each internal terminal **561** is electrically coupled to the resonator element **55** via a conductive bonding member B**4**. The terminals **561** and **564** are electrically coupled as appropriate through internal wiring (not shown) formed in the base **52** to electrically couple the resonator element **55** and the external terminal **564**. The inside and the outside of the package **51** are electrically coupled via the external terminal **564**.

[0040] The resonator element **55** is an AT cut quartz crystal resonator element. Alternatively, the resonator element **55** may not be the AT cut quartz crystal resonator element, but may be, for example, an SC cut quartz crystal resonator element, a BT cut quartz crystal resonator element, a tuning fork type quartz crystal resonator element, a surface acoustic wave resonator, another piezoelectric resonator element, or a MEMS resonator element.

[0041] As shown in FIG. **4**, the resonator **5** is fixed to the bottom surface of the lower recess **212** via the conductive bonding member B**1**. The external terminal **564** and the internal terminal **243** are electrically coupled via the bonding member B**1**.

1-2. Functional Configuration of Oscillator

[0042] FIG. **5** is a functional block diagram of the oscillator **1** of a first embodiment. In FIG. **5**, the same component elements as those in FIGS. **1** to **4** have the same signs. As shown in FIG. **5**, the oscillator **1** of the first embodiment includes the control IC **4**, the resonator **5**, the resonator element **6**, the heat generation IC **7**, and the oscillation IC **8**.

[0043] The oscillation IC **8** includes an oscillation circuit **81** and a temperature sensor **82**, and is supplied with a power supply voltage VOSC from the control IC **4** and operates. The oscillation circuit **81** is a circuit that is electrically coupled to both ends of the resonator element **6**, amplifies the output signal of the resonator element **6**, and feeds back the amplified signal to the resonator element **6**, and thereby, oscillates the resonator element **6** and outputs a reference clock signal CLKIN as a clock signal based on the oscillation signal. In other words, the oscillation circuit **81** is a circuit that operates at a frequency fCLKIN of the reference clock signal CLKIN. For example, the oscillation circuit **81** may be an oscillation circuit using an inverter as an amplification element or may be an oscillation circuit using a bipolar transistor as the amplification element. The reference clock signal CLKIN output from the oscillation circuit **81** is input to the control IC **4**.

[0044] The temperature sensor **82** is a thermosensitive device that detects a temperature and outputs a temperature detection signal TS**1** having a voltage level corresponding to the detected temperature. The temperature sensor **82** is provided in the oscillation IC **8** to detect the temperature of the oscillation IC **8**. The temperature detection signal TS**1** output from the temperature sensor **82** is input to the control IC **4**. The temperature sensor **82** may be, for example, a sensor utilizing temperature dependence of a forward voltage of a PN junction of a diode.

[0045] The heat generation IC **7** includes a temperature control element **71** and a temperature sensor **72**. The temperature control element **71** is an element that controls a temperature of the resonator element **6** based on a temperature control signal OVC output from the control IC **4**, and may be a heat generation element. For example, the temperature control element **71** is a CMOS transistor, and an amount of generated heat changes according to a voltage of the temperature control signal OVC input to a gate. The larger the amount of heat generated by the temperature control element **71**, the higher the temperature of the resonator element **6**. The amount of heat generated by the temperature control element **71** is controlled by the control IC **4** so that the temperature of the resonator element **6** is constant at a target set temperature. For example, the set temperature may be a fixed value such as 80° C., or may be optionally set within a predetermined range such as a range from 70°° C. to 125° C.

[0046] The temperature sensor **72** is a thermosensitive device that detects a temperature and outputs a temperature detection signal TS**2** having a voltage level corresponding to the detected temperature. The temperature sensor **72** is provided in the heat generation IC **7** to detect the temperature of the heat generation IC **7**. Since the temperature control element **71** is also provided in the heat generation IC **7**, the temperature sensor **72** detects the temperature of the temperature control element **71**. The temperature detection signal TS**2** output from the temperature sensor **72** is input to the control IC **4**. The temperature sensor **72** may be, for example, a sensor utilizing temperature dependence of a forward voltage of a PN junction of a diode.

[0047] As shown in FIG. **3**, the resonator element **6**, the heat generation IC **7**, and the oscillation IC **8** are housed in the inner package **3**, and heat generation of the heat generation IC **7** is controlled by the control IC **4** so that the temperature of the resonator element **6** is kept constant. The heat generation IC **7** is a heat source, radiant heat from the heat generation IC **7** is transmitted to the resonator element **6** and the oscillation IC **8**, and thus a difference occurs between the temperatures of the resonator element **6** and the oscillation IC **8** and the temperature of the heat generation IC **7**. Meanwhile, the resonator element **6** and the oscillation IC **8** are disposed apart from the heat generation IC **7**, and assuming that the thermal distance between the heat generation IC **7** and the resonator element **6** is substantially the same as the thermal distance between the heat generation IC **7** and the oscillation IC **8**, the temperature of the oscillation IC **8** may be regarded as being close to the temperature of the resonator element **6**. That is, the temperature detected by the temperature sensor **82** provided in the oscillation IC **8** is closer to the temperature of the resonator element **6** than the temperature detected by the temperature sensor **72** provided in the heat generation IC **7**. Therefore, as will be described later, the control IC **4** controls heat generation of the heat generation IC **7** based on the temperature detection signal TS**1** output from the temperature sensor **82**. However, depending on the arrangement of the resonator element **6**, the heat generation IC **7**, and the oscillation IC **8**, the temperature of the heat generation IC **7** may be closer to the temperature of the resonator element **6** than the temperature of the oscillation IC **8**, and in this case, the control IC **4** may control the heat generation of the heat generation IC **7** based on the temperature detection signal TS**2** output from the temperature sensor **72**.

[0048] The control IC **4** is externally supplied with a first power supply voltage VDD**1**, a second power supply voltage VDD**2**, and a ground voltage VSS and operates. The control IC **4** includes a microcontroller **40**, a selector **41**, a temperature sensor **42**, an A/D converter circuit **43**, a D/A converter circuit **44**, a fractional N-PLL circuit **45**, a PLL circuit **46**, a switch circuit **47**, an interface circuit **49**, LDO regulators **61** to **67**, a memory **90**, and a register **94**. The PLL is an abbreviation for Phase Locked Loop. The LDO is an abbreviation for Low Drop Out.

[0049] The LDO regulator **61** is a power supply circuit that generates a power supply voltage VOSC as a fixed voltage lower than the first power supply voltage VDD**1** based on the first power supply voltage VDD**1** supplied from the outside of the oscillator **1**. The power supply voltage VOSC is supplied to the oscillation circuit **81** of the oscillation IC **8**.

[0050] The LDO regulator **62** is a power supply circuit that generates a power supply voltage VPFD as a fixed voltage lower than the first power supply voltage VDD**1** based on the first power supply voltage VDD**1**. The power supply voltage VPFD is supplied to the fractional N-PLL circuit **45**.

[0051] The LDO regulator **63** is a power supply circuit that generates a power supply voltage VCP as a fixed voltage lower than the second power supply voltage VDD**2** based on the second power supply voltage VDD**2** supplied from the outside of the oscillator **1**. The power supply voltage VCP is supplied to the fractional N-PLL circuit **45**.

[0052] The LDO regulator **64** is a power supply circuit that generates a power supply voltage VBUF as a fixed voltage lower than the second power supply voltage VDD**2** based on the second power supply voltage VDD**2**. The power supply voltage VBUF is supplied to the fractional N-PLL circuit **45**.

[0053] The LDO regulator **65** is a power supply circuit that generates a power supply voltage VDIV as a fixed voltage lower than the second power supply voltage VDD**2** based on the second power supply voltage VDD**2**. The power supply voltage VDIV is supplied to the fractional N-PLL circuit **45**.

[0054] The LDO regulator **66** is a power supply circuit that generates a power supply voltage VVCO as a fixed voltage lower than the second power supply voltage VDD**2** based on the second power supply voltage VDD**2**. The power supply voltage VVCO is supplied to the PLL circuit **46**.

[0055] The LDO regulator **67** is a power supply circuit that generates a power supply voltage VPLL as a fixed voltage lower than the second power supply voltage VDD**2** based on the second power supply voltage VDD**2**. The power supply voltage VPLL is supplied to the PLL circuit **46**.

[0056] The frequency f.sub.CLKIN of the reference clock signal CLKIN output from the oscillation IC **8** is input, and the fractional N-PLL circuit **45** generates and outputs a clock signal CK**1** obtained by conversion of the frequency f.sub.CLKIN Of the reference clock signal CLKIN into a frequency f.sub.CK1 according to a division ratio commanded by a division ratio control signal DIVC.

[0057] FIG. **6** shows a configuration example of the fractional N-PLL circuit **45**. As shown in FIG. **6**, the fractional N-PLL circuit **45** includes a phase comparator **111**, a charge pump **112**, a low-pass filter **113**, a voltage controlled oscillation circuit **114**, a buffer circuit **115**, and a frequency division circuit **116**.

[0058] The phase comparator **111** is supplied with the power supply voltage VPFD and operates, and the charge pump **112** is supplied with the power supply voltage VCP and operates. The low-pass filter **113**, the voltage controlled oscillation circuit **114**, and the buffer circuit **115** are supplied with the power supply voltage VBUF and operate. The frequency division circuit **116** is supplied with the power supply voltage VDIV and operates.

[0059] The phase comparator **111** compares the phase of the reference clock signal CLKIN with the phase of a clock signal FBCLK output by the frequency division circuit **116**, and outputs a comparison result as a pulse voltage.

[0060] The charge pump **112** converts the pulse voltage output by the phase comparator **111** into a current, and the low-pass filter **113** smooths and converts the current output by the charge pump **112** into a voltage.

[0061] The voltage controlled oscillation circuit **114** outputs an oscillation signal having a frequency that changes according to the output voltage of the low-pass filter **113**. The voltage controlled oscillation circuit **114** can be implemented as various types of oscillation circuits including an LC oscillation circuit formed using an inductance element such as a coil and a capacitance element such as a capacitor and an oscillation circuit using a piezoelectric resonator such as a quartz crystal resonator.

[0062] The buffer circuit **115** buffers the oscillation signal output from the voltage controlled oscillation circuit **114** and outputs the clock signal CK**1**.

[0063] The frequency division circuit **116** outputs the clock signal FBCLK obtained by division of the clock signal CK**1** output by the buffer circuit **115** using a value of the division ratio control signal DIVC as the division ratio.

[0064] The fractional N-PLL circuit **45** having the above-described configuration generates the clock signal CK**1** by feedback control such that the phase of the reference clock signal CLKIN coincides with the phase of a signal obtained by division of the clock signal CK**1** using the division ratio designated by the division ratio control signal DIVC. The division ratio control signal DIVC is delta-sigma modulated, and the division ratio designated by the division ratio control signal DIVC is changed among a plurality of integer division ratios and averaged to be a fractional division ratio. The frequency f.sub.CK1 is a non-integral multiple of the frequency f.sub.CLKIN.

The fractional N-PLL circuit **45** may output the clock signal CK**1** having the frequency f.sub.CK1 that is different from the frequency f.sub.CLKIN and is substantially constant regardless of an outside air temperature according to the division ratio control signal DIVC.
[0065] Returning to description of FIG. **5**, the clock signal CK**1** output from the fractional N-PLL circuit **45** is input, and the PLL circuit **46** generates and outputs a clock signal CK**2** having a frequency f.sub.CK2 that is the same frequency as the frequency f.sub.CK1 of the clock signal CK**1**.
[0066] FIG. **7** shows a configuration example of the PLL circuit **46**. As shown in FIG. **7**, the PLL circuit **46** includes a phase comparator **121**, a charge pump **122**, a low-pass filter **123**, an oscillation circuit **124**, and a buffer circuit **125**.
[0067] The phase comparator **121**, the charge pump **122**, the low-pass filter **123**, and the buffer circuit **125** are supplied with the power supply voltage VPLL and operate. The oscillation circuit **124** is supplied with the power supply voltage VVCO and operates.
[0068] The phase comparator **121** compares the phase of the clock signal CK**1** with phase of the clock signal CK**2** output by the buffer circuit **125**, and outputs a comparison result as a pulse voltage.
[0069] The charge pump **122** converts the pulse voltage output by the phase comparator **121** into a current, and the low-pass filter **123** smooths and converts the current output by the charge pump **122** into a voltage.
[0070] The oscillation circuit **124** generates an oscillation signal VO_XI having a frequency f.sub.CK2 that changes according to the output voltage of the low-pass filter **123**. Specifically, the oscillation circuit **124** is coupled to the resonator **5**, oscillates the resonator **5**, and generates the oscillation signal VO_XI having the frequency corresponding to the output voltage of the low-pass filter **123**. That is, the oscillation circuit **124** outputs an oscillation signal VO_XO obtained by amplifying the oscillation signal VO_XI output from the resonator **5** to the resonator **5**, and thereby, the resonator **5** continues to oscillate.
[0071] The buffer circuit **125** buffers the oscillation signal VO_XI generated by the oscillation circuit **124** and outputs the clock signal CK**2** having the frequency fck**2**. As will be described later, in the normal operation of the oscillator **1**, the clock signal CK**2** is selected as an output clock signal OUT, and the buffer circuit **125** outputs the output clock signal OUT.
[0072] Thus, the resonator **5** and the oscillation circuit **124** form a voltage controlled oscillator **9** having an oscillation frequency that changes according to the output voltage of the low-pass filter **123**.
[0073] The clock signal CK**1** output from the fractional N-PLL circuit **45** is input and the PLL circuit **46** having the above-described configuration synchronizes the phase of the clock signal CK**2** with that of the clock signal CK**1**. That is, the PLL circuit **46** generates and outputs the clock signal CK**2** having the frequency f.sub.CK2 that is the same frequency as the frequency f.sub.CK1 of the clock signal CK**1** by feedback control of the output voltage of the low-pass filter **123** such that the phase of the clock signal CK**1** coincides with the phase of the clock signal CK**2**.
[0074] Returning to the description of FIG. **5**, the clock signal CK**1** output from the fractional N-PLL circuit **45** has the frequency f.sub.CK1 that is a non-integral multiple of the frequency f.sub.CLKIN of the reference clock signal CLKIN, and has larger jitter. In contrast, the clock signal CK**2** output from the PLL circuit **46** has the frequency f.sub.CK2 that is the same as the frequency f.sub.CK1 of the clock signal CK**1** and is generated by oscillation of the resonator **5** having higher frequency stability, and has jitter smaller than that of the clock signal CK**1**.
[0075] The switch circuit **47** outputs the output clock signal OUT obtained by selecting the clock signal CK**1** or the clock signal CK**2** according to a logic level of a switch control signal SWC output from the register **94**. The output clock signal OUT is output to the outside of the oscillator **1**. The output clock signal OUT may be supplied to an external apparatus **100** or may be supplied to an apparatus different from the external apparatus **100**. For example, the clock signal CK**2** with the

smaller jitter may be selected as the output clock signal OUT in the normal operation of the oscillator **1**, and the clock signal CK**1** may be selected as the output clock signal OUT in the inspection of the clock signal CK**1**.

[0076] The temperature sensor **42** is a thermosensitive device that detects a temperature and outputs a temperature detection signal TS**3** having a voltage level corresponding to the detected temperature. The temperature sensor **42** is provided in the control IC **4** to detect the temperature of the control IC **4**. As shown in FIG. **1**, the control IC **4** is disposed close to the outer lid **22**, the distance between the resonator element **6** and the temperature sensor **42** is larger than the distance between the resonator element **6** and the temperature sensor **82** provided in the oscillation IC **8**, and the temperature of the control IC **4** is easily affected by the outside air temperature of the oscillator **1**. Therefore, when the amount of heat generated by the control IC **4** is substantially constant, the temperature sensor **42** can detect changes of the outside air temperature of the oscillator **1**. The temperature detection signal TS**2** output from the temperature sensor **72** is input to the control IC **4**. The temperature sensor **72** may be, for example, a sensor utilizing temperature dependence of a forward voltage of a PN junction of a diode.

[0077] The selector **41** selects and outputs one of the temperature detection signal TS**1** output from the oscillation IC **8**, the temperature detection signal TS**2** output from the heat generation IC **7**, and the temperature detection signal TS**3** output from the temperature sensor **42**. In the embodiment, the selector **41** selects the temperature detection signals TS**1**, TS**2**, and TS**3** in a time-division manner and periodically outputs the selected signals.

[0078] The A/D converter circuit **43** converts the respective voltages of the temperature detection signals TS**1**, TS**2**, and TS**3** as analog signals output from the selector **41** in the time-division manner into temperature codes DTS**1**, DTS**2**, and DTS**3** as digital signals, respectively. The A/D converter circuit **43** may convert the temperature detection signals TS**1**, TS**2**, and TS**3** into the temperature codes DTS**1**, DTS**2**, and DTS**3** after converting voltage levels by resistance voltage division or the like.

[0079] The microcontroller **40** includes a CPU **10** and a memory **15**. The CPU is an abbreviation for Central

[0080] Processing Unit. Temperature control data **91** and temperature compensation data **92** are stored in the nonvolatile memory **90** and are transferred to the memory **15** when the oscillator **1** is started up. Further, a temperature control program and a temperature compensation program (not shown) are stored in the nonvolatile memory **90** and are transferred to the memory **15** when the oscillator **1** is started up.

[0081] The CPU **10** functions as a temperature control circuit **11** by executing the temperature control program transferred to the memory **15**. The temperature control circuit **11** controls the operation of the temperature control element **71** provided in the heat generation IC **7**. Specifically, the temperature control circuit **11** outputs a temperature control code DOVC for controlling the amount of heat generated by the temperature control element **71** based on the temperature code DTS**1** and the temperature control data **91** transferred to and stored in the memory **15**. For example, the temperature control data **91** may include information on the target set temperature for the temperature of the resonator element **6** and gain information for controlling the amount of heat generated by the temperature control element **71**. Alternatively, when the target set temperature for the temperature of the resonator element **6** varies depending on the outside air temperature, the temperature control data **91** may include information indicating a relationship between the temperature code DTS**3** and the set temperature. In this case, the temperature control circuit **11** outputs the temperature control code DOVC based on the temperature codes DTS**1** and DTS**3**, and the temperature control data **91**.

[0082] The CPU **10** functions as a temperature compensation circuit **12** by executing the temperature compensation program transferred to the memory **15**. The temperature compensation circuit **12** performs temperature compensation on the frequency of the reference clock signal

CLKIN generated by the oscillation circuit **81** provided in the oscillation IC **8** oscillating the resonator element **6**. Specifically, the temperature compensation circuit **12** outputs the division ratio control signal DIVC for causing the fractional N-PLL circuit **45** to output the clock signal CK**1** having a constant frequency regardless of the temperature based on the temperature code DTS**3** and the temperature compensation data **92** transferred to the memory **15**. For example, the temperature compensation data **92** may be table information indicating a relationship between the temperature code DTS**3** and the frequency of the reference clock signal CLKIN or may be information on a coefficient value for each order in a mathematical formula indicating the relationship. Alternatively, the temperature compensation data **92** may be information indicating a relationship between the temperature code DTS**3** and a value of a fractional division ratio of the fractional N-PLL circuit **45** calculated from the relationship between the temperature code DTS**3** and the frequency of the reference clock signal CLKIN.

[0083] The D/A converter circuit **44** converts the temperature control code DOVC as a digital signal output from the temperature control circuit **11** into the temperature control signal OVC as an analog signal. The temperature control signal OVC is supplied to the temperature control element **71** of the heat generation IC **7**.

[0084] The interface circuit **49** is a circuit for data communication with the external apparatus **100** coupled to the oscillator **1**. Specifically, the interface circuit **49** writes or reads data to or from the memory **90**, the register **94**, or the memory **15** of the microcontroller **40** in response to a request from the external apparatus **100**. The interface circuit **49** may be, for example, an interface circuit corresponding to an I.sup.2C bus or an interface circuit corresponding to an SPI bus. The I.sup.2C is an abbreviation for Inter-Integrated Circuit. The SPI is an abbreviation for Serial Peripheral Interface.

[0085] In an inspection process at the time of manufacturing of the oscillator **1**, an inspection apparatus as the external apparatus **100** may set the switch control signal SWC for causing the switch circuit **47** to select the clock signal CK**1** via the interface circuit **49** and inspect the clock signal CK**1**. The inspection apparatus as the external apparatus **100** writes the temperature control data **91** and the temperature compensation data **92**, and further writes the temperature control program and the temperature compensation program in the memory **90** via the interface circuit **49**. Note that the external apparatus **100** may set the temperature control data **91** and the temperature compensation data **92** in the register **94** when the oscillator **1** is started up.

1-3. Layout Configuration of Control IC

[0086] As described above, the frequency f.sub.CK1 of the clock signal CK**1** is a non-integral multiple of the frequency f.sub.CLKIN of the reference clock signal CLKIN, and the frequency f.sub.CK2 of the clock signal CK**2** is the same as the frequency f.sub.CK1 the clock signal CK**1**. In the normal operation, the clock signal CK**2** is selected as the output clock signal OUT, and the frequency f.sub.CLKIN of the reference clock signal CLKIN input to the control IC **4** and the frequency f.sub.CK2 of the output clock signal OUT output from the control IC **4** are different from each other. Accordingly, the signal at the frequency f.sub.CLKIN and the signal at the frequency f.sub.CK2 may interfere with each other and beat noise having a frequency of the difference between the frequency f.sub.CLKIN and the frequency f.sub.CK2 may be generated as spurious noise. Therefore, in the embodiment, in order to reduce the spurious noise, the layout configuration of the control IC **4** is devised.

[0087] FIG. **8** shows a layout configuration of the control IC **4** in the first embodiment. In FIG. **8**, only the arrangement of some of pads and the like necessary for the description is illustrated. As shown in FIG. **8**, the control IC **4** includes a semiconductor substrate **400**. In a plan view, the semiconductor substrate **400** has a first side **401**, a second side **402** as an opposite side to the first side **401**, a third side **403** intersecting the first side **401** and the second side **402**, and a fourth side **404** as an opposite side to the third side **403**. Note that the contours of the control IC **4** and the semiconductor substrate **400** substantially coincide with each other in the plan view. Accordingly, the control IC **4** has the first side **401**, the second side **402**, the third side **403**, and the fourth side **404** in the plan view.

[0088] On the semiconductor substrate **400**, pads P**1**, P**2**, P**3**, P**4** and P**5** are arranged along the first side **401**. The pad P**1** is a clock input terminal to which the reference clock signal CLKIN is input. The pad P**2** is a ground terminal to which the ground voltage VSS is supplied. The pad P**3** is a power supply voltage input terminal to which the first power supply voltage VDD**1** as the source of the power supply voltage VOSC supplied to the oscillation circuit **81** is input. The pad P**4** is a ground terminal to which the ground voltage VSS is supplied. The pad P**5** is a power supply voltage output terminal that outputs the power supply voltage VOSC. The pad P**1**, the pad P**2**, the pad P**3**, the pad P**4**, and the pad P**5** are arranged in this order along the first side **401**.

[0089] Further, on the semiconductor substrate **400**, pads P**6**, P**7**, P**8**, P**9**, P**10**, and P**11** are arranged along the second side **402**. The pad P**6** is a power supply voltage input terminal to which the second power supply voltage VDD**2** as the source of the power supply voltages VVCO and VPLL supplied to the PLL circuit **46** is input. The pad P**7** is a ground terminal to which the ground voltage VSS is supplied. The pad P**8** is a clock output terminal from which the output clock signal OUT having the frequency different from that of the reference clock signal CLKIN is output. The pad P**9** is a ground terminal to which the ground voltage VSS is supplied. The pad P**10** is a first resonator terminal coupled to the resonator **5**, the oscillation circuit **124**, and the buffer circuit **125**. The pad P**11** is a second resonator terminal coupled to the resonator **5** and the oscillation circuit **124**. That is, the oscillation signal VO_XI output from the resonator **5** is input to the oscillation circuit **124** and the buffer circuit **125** via the pad P**10** as the first resonator terminal, and the oscillation signal VO_XO amplified by the oscillation circuit **124** is input to the resonator **5** via the pad P**11** as the second resonator terminal. The pad P**6**, the pad P**7**, the pad P**8**, the pad P**9**, the pad P**10**, and the pad P**11** are arranged in this order along the second side **402**.

[0090] As described above, the pad P**1** to which the reference clock signal CLKIN having the frequency f.sub.CLKIN is input and the pad P**8** to which the output clock signal OUT having the frequency f.sub.CK2 is output are provided in positions apart from each other. Further, the fractional N-PLL circuit **45** to which the reference clock signal CLKIN having the frequency f.sub.CLKIN is input is provided in a position closer to the first side **401** than the second side **402**. Furthermore, the PLL circuit **46** that outputs the clock signal CK**2** having the frequency f.sub.CK2 is provided in a position closer to the second side **402** than the first side **401**. The first power supply voltage VDD**1** as the source of the power supply voltage VOSC supplied to the oscillation circuit **81** operating at the frequency f.sub.CLKIN and the second power supply voltage VDD**2** as the source of the power supply voltages VVCO and VPLL supplied to the PLL circuit **46** operating at the frequency f.sub.CK2 are input from the two different pads P**2** and P**5**, respectively, and the pad P**2** and the pad P**5** are provided in positions apart from each other. That is, since the arrangement region of the circuit that operates at the frequency f.sub.CLKIN based on the first power supply voltage VDD**1** and the arrangement region of the circuit that operates at the frequency f.sub.CK2 based on the second power supply voltage VDD**2** are separated as much as possible, spurious noise generated due to the interference between the signal having the frequency f.sub.CLKIN and the signal having frequency f.sub.CK2 is reduced.

[0091] Further, the pad P**2** as the ground terminal is disposed between the pad Pl as the clock input terminal and the pad P**3** as the power supply voltage input terminal along the first side **401**. That is, since the pad P**2** as the ground terminal is disposed next to the pad P**1** as the clock input terminal, as shown in FIG. **9**, the magnetic field generated by a current I**1** based on the charging and discharging when the reference clock signal CLKIN is input to the pad P**1** is weakened by the magnetic field generated by a current I**2** flowing to the ground via the pad P**2**. Therefore, the electromagnetic field coupling between the node to which the reference clock signal CLKIN is input and the node of the first power supply voltage VDD**1** becomes smaller, and an induced current I**3** flowing to the node of the first power supply voltage VDD**1** is reduced by electromagnetic induction based on the magnetic field, and thereby, fluctuations of the first power supply voltage VDD**1** are reduced. Further, the capacitive coupling between the pad P**1** and the pad P**3** is reduced by the pad P**2** as the ground terminal, and thereby, noise of the frequency f.sub.CLKIN based on the reference clock signal CLKIN is less likely to be superimposed on the first power supply voltage VDD**1**.

[0092] The pads P**2** and P**4** as the ground terminals are disposed between the pad P**5** as the power supply voltage output terminal and the pad P**1** as the clock input terminal along the first side **401**. Accordingly, the magnetic field generated by the current based on charging and discharging when the reference clock signal CLKIN is input to the pad P**1** is weakened by the magnetic field generated by currents flowing to the ground via the pads P**2** and P**4**. Therefore, the electromagnetic field coupling between the node to which the reference clock signal CLKIN is input and the node of the power supply voltage VOSC becomes smaller, and an induced current flowing to the node of the power supply voltage VOSC is reduced by electromagnetic induction based on the magnetic field, and thereby, fluctuations of the power supply voltage VOSC are reduced.

[0093] The pad P**4** as the ground terminal is disposed between the pad P**5** as the power supply voltage output terminal and the pad P**3** as the power supply voltage input terminal along the first side **401**. Accordingly, the magnetic field generated by the current based on the fluctuations of the power supply voltage VOSC and the magnetic field generated by the current based on the fluctuations of the first power supply voltage VDD**1** are weakened by the magnetic field generated by the current flowing to the ground. Therefore, the electromagnetic field coupling between the node of the first power supply voltage VDD**1** and the node of the power supply voltage VOSC becomes smaller, and an induced current flowing to the node of the first power supply voltage VDD**1** and the node of the power supply voltage VOSC is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the power supply voltage VOSC and fluctuations of the first power supply voltage VDD**1** are reduced.

[0094] Since the pads P**2** and P**4** as the ground terminals are disposed on both sides of the pad P**3** as the power supply voltage input terminal along the first side **401**, the magnetic field generated by the current based on the fluctuations of the first power supply voltage VDD**1** is weakened by the magnetic field generated by the current flowing to the ground. Therefore, the electromagnetic field coupling between the node of the first power supply voltage VDD**1** and each node inside the control IC **4** is reduced, and fluctuations of each node are reduced.

[0095] Further, the pad P**7** as the ground terminal is disposed between the pad P**8** as the clock output terminal and the pad P**6** as the second power supply voltage input terminal along the second side **402**. That is, since the pad P**7** as the ground terminal is disposed next to the pad P**8** as the clock output terminal, the magnetic field generated by the current based on charging and discharging when the output clock signal OUT is output from the pad P**8** is weakened by the magnetic field generated by the current flowing to the ground via the pad P**7**. Therefore, the electromagnetic field coupling between the node from which the output clock signal OUT is output and the node of the second power supply voltage VDD**2** becomes smaller, and an induced current flowing to the node of the second power supply voltage VDD**2** is reduced by the electromagnetic induction based on the magnetic field, and the noise of the frequency f.sub.CK2 superimposed on the second power supply voltage VDD**2** is reduced. Further, the capacitive coupling between the pad P**6** and the pad P**8** is reduced by the pad P**7** as the ground terminal, and thereby, noise of the frequency fcx**2** based on the output clock signal OUT is less likely to be superimposed on the second power supply voltage VDD**2**.

[0096] The pad P**9** as the ground terminal is disposed between the pad P**10** as the first resonator terminal and the pad P**8** as the clock output terminal along the second side **402**. Accordingly, the magnetic field generated by the current based on charging and discharging when the output clock signal OUT is output from the pad P**8** is weakened by the magnetic field generated by the current flowing to the ground via the pad P**9**. Therefore, the electromagnetic field coupling between the

node to which the output clock signal OUT is output and the node to which the oscillation signal VO_XI is input becomes smaller, and an induced current flowing to the node to which the VO_XI is input is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the oscillation signal VO_XI are reduced.
[0097] The pads P**7** and P**9** as the ground terminals are disposed between the pad P**10** as the first resonator terminal and the pad P**6** as the power supply voltage input terminal along the second side **402**. Accordingly, the magnetic field generated by the current based on the fluctuations of the oscillation signal VO_XI and the magnetic field generated by the current based on the fluctuations of the second power supply voltage VDD**2** are weakened by the magnetic field generated by the current flowing to the ground. Therefore, the electromagnetic field coupling between the node of the second power supply voltage VDD**2** and the node to which the oscillation signal VO_XI is input becomes smaller, and an induced current flowing to the node of the second power supply voltage VDD**2** and the node to which the oscillation signal VO_XI is input is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the oscillation signal VO_XI are reduced and fluctuations of the second power supply voltage VDD**2** are reduced.
[0098] Since the pad P**7** as the ground terminal is disposed next to the pad P**6** as the power supply voltage input terminal along the second side **402**, the magnetic field generated by the current based on the fluctuations of the second power supply voltage VDD**2** is weakened by the magnetic field generated by the current flowing to the ground. Therefore, the electromagnetic field coupling between the node of the second power supply voltage VDD**2** and each node in the control IC **4** is reduced, and fluctuations of each node is reduced.
[0099] As shown in FIG. **8**, the LDO regulator **61** that outputs the power supply voltage VOSC is provided in a position close to the pad P**5**. Accordingly, a wire **411** from the output of the LDO regulator **61** to the pad P**5** is shorter, impedance thereof is reduced, and noise propagating from the node of the first power supply voltage VDD**1** to the node of the power supply voltage VOSC is reduced by inverse PSNR characteristics of the LDO regulator **61**.
[0100] The LDO regulator **62** that outputs the power supply voltage VPFD, the LDO regulator **63** that outputs the power supply voltage VCP, the LDO regulator **64** that outputs the power supply voltage VBUF, and the LDO regulator **65** that outputs the power supply voltage VDIV are provided in positions close to the fractional N-PLL circuit **45**. Accordingly, each wire from each output of the LDO regulators **62**, **63**, **64**, and **65** to the fractional N-PLL circuit **45** is shorter, impedance thereof is reduced, and noise propagating from the node of the first power supply voltage VDD**1** to each node of the power supply voltage VPFD, the power supply voltage VCP, the power supply voltage VBUF, and the power supply voltage VDIV is reduced by inverse PSNR characteristics of the LDO regulators **62**, **63**, **64**, and **65**.
[0101] The LDO regulator **66** that outputs the power supply voltage VVCO and the LDO regulator **67** that outputs the power supply voltage VPLL are provided in positions close to the PLL circuit **46**. Accordingly, each wire from each output of the LDO regulators **66** and **67** to the PLL circuit **46** is shorter, impedance thereof is reduced, and noise propagating from the node of the second power supply voltage VDD**2** to each node of the power supply voltage VVCO, the power supply voltage VPLL is reduced by inverse PSNR characteristics of the LDO regulators **66** and **67**.
[0102] As shown in FIG. **8**, a first power supply wire **410** is provided on the semiconductor substrate **400**. The first power supply wire **410** is coupled to the pad P**3** and the LDO regulators **61** and **62**, and the first power supply voltage VDD**1** is supplied from the pad P**3** to the LDO regulators **61** and **62** via the first power supply wire **410**. Further, the first power supply wire **410** extends along a part of the outer edge of the arrangement region of the fractional N-PLL circuit **45**.
[0103] A second power supply wire **420** is provided on the semiconductor substrate **400**. The second power supply wire **420** is coupled to the pad P**6** and the LDO regulators **63**, **64**, **65**, **66**, and **67**, and the second power supply voltage VDD**2** is supplied from the pad P**6** to the LDO regulators **63**, **64**, **65**, **66**, and **67** via the second power supply wire **420**. The second power supply wire **420** is

provided along a part of the outer edge of the arrangement region of the PLL circuit **46** and a part of the outer edge of the arrangement region of the fractional N-PLL circuit **45**.

[0104] As shown in FIG. **8**, a part of the first power supply wire **410** and a part of the second power supply wire **420** are parallel to each other, and fluctuations of the first power supply voltage VDD**1** and fluctuations of the second power supply voltage VDD**2** are likely to interfere with each other by capacitive coupling between the first power supply wire **410** and the second power supply wire **420**. Accordingly, a shield wire **430** is provided between the first power supply wire **410** and the second power supply wire **420**. For example, since the first power supply wire **410**, the second power supply wire **420**, and the shield wire **430** are provided in the same wiring layer, capacitive coupling between the first power supply wire **410** and the second power supply wire **420** is effectively suppressed. Further, the first power supply wire **410** and the second power supply wire **420** are not annular and disposed so as not to come close to each other except a part with the shield wire **430** in between for reduction of the capacitive coupling between the first power supply wire **410** and the second power supply wire **420**.

[0105] The pads P**10** and P**11** are pads coupled to the resonator **5**, and each node coupled to each of the pads P**10** and P**11** is a high-impedance node, and noise is easily superimposed thereon. In particular, the pad P**10** is a pad to which the oscillation signal VO_XI output from the resonator **5** is input, and the noise superimposed on the node of the pad P**10** is amplified by the oscillation circuit **124**. Therefore, the first power supply wire **410** and the second power supply wire **420** are provided farther from the pads P**10** and P**11**. As a result, the shortest distance between the pad P**10** and the second power supply wire **420** is longer than the shortest distance between the pad P**8** as the clock output terminal and the second power supply wire **420**.

[0106] The control IC **4** is an example of “integrated circuit device”, and the oscillation IC **8** is an example of “second integrated circuit device”. The oscillation circuit **81** is an example of “first circuit”, and the PLL circuit **46** is an example of “second circuit”. The pad P**2** is an example of “first ground terminal”, the pad P**7** is an example of “second ground terminal”, the pad P**4** is an example of “third ground terminal”, and the pad P**9** is an example of “fourth ground terminal”. The pad P**3** is an example of “first power supply voltage input terminal”, and the pad P**6** is an example of “second power supply voltage input terminal”.

1-4. Functions and Effects

[0107] As described above, in the oscillator **1** of the first embodiment, the control IC **4** has the layout configuration shown in FIG. **8**, and thereby, the electromagnetic field coupling and capacitive coupling between the nodes are reduced and the spurious noise superimposed on the output clock signal OUT is reduced. According to the oscillator **1** of the first embodiment, the output clock signal OUT having a higher S/N ratio can be output. Further, since the plurality of wires outside the control IC **4** respectively coupled to the pads P**1** to P**11** are also arranged in the same manner as the pads P**1** to P**11**, the above-described effects can also be obtained outside the control IC **4**.

2. Second Embodiment

[0108] Regarding a second embodiment, the same configurations as those of the first embodiment have the same signs, and the same description as that of the first embodiment will be omitted or simplified and differences from the first embodiment will be mainly described as below.

[0109] A structure of an oscillator **1** of the second embodiment is the same as that in FIGS. **1** to **4**, and the illustration and description thereof are omitted.

[0110] In the oscillator **1** of the first embodiment, in the control IC **4**, the pad P**4** as the ground terminal is disposed next to the pad P**1** to which the reference clock signal CLKIN is input, and the pads P**7** and P**9** as the ground terminals are disposed on both sides of the pad P**8** to which the output clock signal OUT is output. Accordingly, as shown in FIG. **10**, when the impedance of the ground is higher, slight spurious noise may be superimposed on the output clock signal OUT due to capacitive coupling between the pad P**1** and the ground and capacitive coupling between the ground and the pad P**8**.

[0111] In the oscillator **1** of the second embodiment, in the control IC **4**, the pad P**2** is coupled to a first ground outside the control IC **4** and the pads P**7** and P**9** are coupled to a second ground outside the control IC **4** different from the first ground. FIG. **11** shows a layout configuration of the control IC **4** in the second embodiment. In FIG. **11**, the same component elements as those in FIG. **8** have the same signs.

[0112] The layout shown in FIG. **11** is different from the layout in FIG. **8** in that a first ground voltage VSS**1** is supplied to the pads P**2** and P**4** and a second ground voltage VSS**2** is supplied to the pads P**7** and P**9**, and the rest of the layout is the same as the layout in FIG. **8**. The pads P**2** and P**4** are coupled to the first ground, and the pads P**7** and P**9** are coupled to the second ground different from the first ground. The first ground voltage VSS**1** is the voltage of the first ground, and the second ground voltage VSS**2** is the voltage of the second ground. Accordingly, as shown in FIG. **12**, capacitive coupling between the pad P**1** and the first ground coupled to the adjacent pad P**2** and capacitive coupling between the pad P**8** and the second ground coupled to the pads P**7** and P**9** on both sides thereof are generated, however, the first ground and the second ground are separated and spurious noise is not superimposed on the output clock signal OUT by these capacitive couplings.

[0113] The functional block diagram of the oscillator **1** of the second embodiment is the same as that in FIG. **5** except that the first ground voltage VSS**1** and the second ground voltage VSS**2** are supplied, and the illustration and description thereof are omitted.

[0114] As described above, according to the oscillator **1** of the second embodiment, like the oscillator **1** of the first embodiment, the electromagnetic field coupling and capacitive coupling between the nodes are reduced, and the spurious noise superimposed on the output clock signal OUT is reduced. Further, since the first ground coupled to the pad P**2** and the second ground coupled to the pads P**7** and P**9** are separated from each other, the interference between the reference clock signal CLKIN input to the pad P**1** and the output clock signal OUT output from the pad P**8** is reduced, and the spurious noise superimposed on the output clock signal OUT is further reduced. According to the oscillator **1** of the first embodiment, the output clock signal OUT having a higher S/N ratio can be output. Further, since the plurality of wires outside the control IC **4** respectively coupled to the pads P**1** to P**11** are also arranged in the same manner as the pads P**1** to P**11**, the above-described effects can also be obtained outside the control IC **4**.

3. Modifications

[0115] The present disclosure is not limited to the embodiments, and various modifications can be made within the scope of the gist of the present disclosure.

[0116] In the above-described respective embodiments, the temperature control element **71** and the temperature sensor **72** are provided in the heat generation IC **7**, however, the temperature control element **71** and the temperature sensor **72** may be provided separately. In the above-described embodiments, the temperature sensor **82** is provided in the oscillation IC **8**, however, the temperature sensor **82** and the oscillation IC **8** may be provided separately. In the above described embodiments, the temperature sensor **42** is provided in the control IC **4**, however, the temperature sensor **42** and the control IC **4** may be provided separately. In these cases, for example, the temperature sensors **72**, **82**, and **42** may be thermistors or platinum resistors.

[0117] In the above-described respective embodiments, the control IC **4** includes the single temperature sensor **42**, however, a plurality of temperature sensors may be provided. In this case, for example, the A/D converter circuit **43** may convert a plurality of temperature detection signals output from the plurality of temperature sensors into a plurality of temperature codes, and the microcontroller **40** may perform temperature control and temperature compensation based on the plurality of temperature codes. For example, the microcontroller **40** may perform temperature control and temperature compensation using an average value of the plurality of temperature codes as the temperature code DTS**3**.

[0118] In the above-described respective embodiments, the single A/D converter circuit **43** converts the voltages of the temperature detection signals TS**1**, TS**2**, and TS**3** into the temperature codes DTS**1**, DTS**2**, and DTS**3** in the time-division manner, respectively, however, for example, the control IC **4** may include a plurality of A/D converter circuits and the plurality of A/D converter circuits may respectively convert the voltages of the temperature detection signals TS**1**, TS**2**, and TS**3** into the temperature codes DTS**1**, DTS**2**, and DTS**3**.
[0119] In the above-described respective embodiments, the temperature control element **71** is the heat generation element such as a CMOS transistor, however, the temperature control element **71** may be an element that can control the temperature of the resonator element **6** and may be a heat absorbing element such as a Peltier element depending on a relationship between the target set temperature of the temperature of the resonator element **6** and the outside air temperature.
[0120] The above-described embodiments and modifications are just examples, and the present disclosure is not limited thereto. For example, the respective embodiments and the respective modifications can be combined as appropriate.
[0121] The present disclosure includes substantially the same configurations as the configurations described in the embodiments, for example, configurations having the same functions, methods, and results or configurations having the same purposes and effects. The present disclosure includes a configuration in which a non-essential portion of the configuration described in the embodiment is replaced. Further, the present disclosure includes a configuration that exerts the same function and effect or a configuration that can achieve the same purpose as the configurations described in the embodiments. Furthermore, the present disclosure includes a configuration with the addition of a known technique to the configuration described in the embodiments.
[0122] The following configurations are derived from the above-described embodiments and the modifications.
[0123] An integrated circuit device according an aspect having a first side and a second side as an opposite side to the first side in a plan view, includes a clock input terminal disposed along the first side, to which a reference clock signal is input, a first power supply voltage input terminal disposed along the first side, to which a first power supply voltage as a source of a power supply voltage supplied to a first circuit that operates at a frequency of the reference clock signal is input, a clock output terminal disposed along the second side, from which an output clock signal having a frequency different from that of the reference clock signal is output, a second power supply voltage input terminal disposed along the second side, to which a second power supply voltage as a source of a power supply voltage supplied to a second circuit that operates at the frequency of the output clock signal is input, a first ground terminal disposed between the clock input terminal and the first power supply voltage input terminal along the first side, and a second ground terminal disposed between the clock output terminal and the second power supply voltage input terminal along the second side.
[0124] In the integrated circuit device, the magnetic field generated by the current based on charging and discharging when the reference clock signal is input to the clock input terminal is weakened by the magnetic field generated by the current flowing to the ground via the first ground terminal. Therefore, the electromagnetic field coupling between the node to which the reference clock signal is input and the node of the first power supply voltage becomes smaller, and an induced current flowing to the node of the first power supply voltage is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the first power supply voltage are reduced. Further, the capacitive coupling between the clock input terminal and the first power supply voltage input terminal is reduced by the first ground terminal, and thereby, noise based on the reference clock signal is less likely to be superimposed on the first power supply voltage.
[0125] In the integrated circuit device, the magnetic field generated by the current based on charging and discharging when the output clock signal is output from the clock output terminal is

weakened by the magnetic field generated by the current flowing to the ground via the second ground terminal. Therefore, the electromagnetic field coupling between the node from which the output clock signal is output and the node of the second power supply voltage becomes smaller, and an induced current flowing to the node of the second power supply voltage is reduced by the electromagnetic induction based on the magnetic field, and the noise superimposed on the second power supply voltage is reduced. Further, the capacitive coupling between the clock output terminal and the second power supply voltage input terminal is reduced by the second ground terminal, and thereby, noise based on the output clock signal is less likely to be superimposed on the second power supply voltage.
[0126] As described above, according to the integrated circuit device, the electromagnetic field coupling and capacitive coupling between the nodes via the terminals are reduced and the spurious noise superimposed on the output clock signal is reduced, and an output clock signal having a higher S/N ratio can be output.
[0127] The integrated circuit device according to the aspect may include a power supply circuit generating the power supply voltage supplied to the first circuit based on the first power supply voltage, a power supply voltage output terminal disposed along the first side and outputting the power supply voltage supplied to the first circuit, and a third ground terminal disposed between the power supply voltage output terminal and the first power supply voltage input terminal or between the power supply voltage output terminal and the clock input terminal.
[0128] In the integrated circuit device, the magnetic field generated by the current based on charging and discharging when the reference clock signal is input to the clock input terminal is weakened by the magnetic field generated by the current flowing to the ground via the third ground terminal. Therefore, the electromagnetic field coupling between the node to which the reference clock signal is input and the node of the power supply voltage supplied to the first circuit becomes smaller, and an induced current flowing to the node of the power supply voltage is reduced by the electromagnetic induction based on the magnetic field and fluctuations of the power supply voltage are reduced. Further, the magnetic field generated by the current based on the fluctuations of the power supply voltage or the magnetic field generated by the current based on the fluctuations of the first power supply voltage are weakened by the magnetic field generated by the current flowing to the ground via the third ground terminal. Accordingly, the electromagnetic field coupling between the node of the first power supply voltage and the node of the power supply voltage supplied to the first circuit becomes smaller, and an induced current flowing to the node of the first power supply voltage and the node of the power supply voltage is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the power supply voltage and fluctuations of the first power supply voltage are reduced. Therefore, according to the integrated circuit device, the electromagnetic field coupling and capacitive coupling between the nodes via the terminals are reduced, and an output clock signal having a higher S/N ratio can be output.
[0129] In the integrated circuit device according to the aspect, the clock input terminal, the first ground terminal, the first power supply voltage input terminal, the third ground terminal, and the power supply voltage output terminal may be arranged in this order along the first side.
[0130] According to the integrated circuit device, the electromagnetic field coupling and capacitive coupling between the nodes via the terminals are reduced, and an output clock signal having a higher S/N ratio can be output.
[0131] The integrated circuit device according to the aspect may include an oscillation circuit oscillating a resonator to generate an oscillation signal, a buffer circuit buffering the oscillation signal and outputting the output clock signal, a first resonator terminal coupled to the resonator and the buffer circuit, and a fourth ground terminal disposed between the first resonator terminal and the clock output terminal or between the first resonator terminal and the second power supply voltage input terminal along the second side.
[0132] According to the integrated circuit device, the magnetic field generated by the current based

on charging and discharging when the output clock signal is output from the clock output terminal is weakened by the magnetic field generated by the current flowing to the ground via the fourth ground terminal. Therefore, the electromagnetic field coupling between the node to which the output clock signal is output and the node coupled to the first resonator terminal becomes smaller, and an induced current flowing to the node coupled to the first resonator terminal is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the oscillation signal based on oscillation of the resonator are reduced. Further, the magnetic field generated by the current based on the fluctuations of the oscillation signal and the magnetic field generated by the current based on the fluctuations of the second power supply voltage are weakened by the magnetic field generated by the current flowing to the ground via the fourth ground terminal. Therefore, the electromagnetic field coupling between the node of the second power supply voltage and the node coupled to the first resonator terminal becomes smaller, and an induced current flowing to the node of the second power supply voltage and the node coupled to the first resonator terminal is reduced by the electromagnetic induction based on the magnetic field, and thereby, fluctuations of the oscillation signal based on oscillation of the resonator and fluctuations of the second power supply voltage are reduced. Therefore, according to the integrated circuit device, the electromagnetic field coupling and capacitive coupling between the nodes via the terminals are reduced, and an output clock signal having a higher S/N ratio can be output.
[0133] In the integrated circuit device according to the aspect, the second power supply voltage input terminal, the second ground terminal, the clock output terminal, the fourth ground terminal, and the first resonator terminal may be arranged in this order along the second side.
[0134] According to the integrated circuit device, the electromagnetic field coupling and capacitive coupling between the nodes via the terminals are reduced, and an output clock signal having a higher S/N ratio can be output.
[0135] The integrated circuit device according to the aspect includes a power supply wire coupled to the second power supply voltage input terminal, wherein a shortest distance between the first resonator terminal and the power supply wire may be longer than a shortest distance between the clock output terminal and the power supply wire.
[0136] According to the integrated circuit device, the node coupled to the first resonator terminal is a high-impedance node, and noise is easily superimposed thereon, however, the power supply wire from which the second power supply voltage is supplied is provided farther from the first resonator terminal, and noise superimposed on the oscillation signal based on oscillation of the resonator is reduced and an output clock signal having a higher S/N ratio can be output.
[0137] The integrated circuit device according to the aspect may include a first power supply wire coupled to the first power supply voltage input terminal, a second power supply wire coupled to the second power supply voltage input terminal, and a shield wire provided between the first power supply wire and the second power supply wire.
[0138] According to the integrated circuit device, capacitive coupling between the first power supply wire and the second power supply wire is effectively suppressed by the shield wire, and thereby, the possibility that fluctuations of the first power supply voltage and fluctuations of the second power supply voltage interfere with each other may be reduced and an output clock signal having a higher S/N ratio can be output.
[0139] In the integrated circuit device according to the aspect, the first ground terminal is coupled to an external first ground, and the second ground terminal is coupled to an external second ground different from the first ground.
[0140] According to the integrated circuit device, the first ground coupled to the first ground terminal and the second ground coupled to the second ground terminal are separated from each other, and thereby, the interference between the reference clock signal input to the clock input terminal and the output clock signal output from the clock output terminal is reduced and the spurious noise superimposed on the output clock signal is further reduced.

[0141] An oscillator according to an aspect includes the integrated circuit device according to the aspect.
[0142] The oscillator according to the aspect may include a resonator element and a second integrated circuit device oscillating the resonator element and outputting the reference clock signal.
[0143] The oscillator according to the aspect may include a temperature control element controlling a temperature of the resonator element.

## Claims

**1**. An integrated circuit device having a first side and a second side as an opposite side to the first side in a plan view, comprising: a clock input terminal disposed along the first side, to which a reference clock signal is input; a first power supply voltage input terminal disposed along the first side, to which a first power supply voltage as a source of a power supply voltage supplied to a first circuit that operates at a frequency of the reference clock signal is input; a clock output terminal disposed along the second side, from which an output clock signal having a frequency different from that of the reference clock signal is output; a second power supply voltage input terminal disposed along the second side, to which a second power supply voltage as a source of a power supply voltage supplied to a second circuit that operates at the frequency of the output clock signal is input; a first ground terminal disposed between the clock input terminal and the first power supply voltage input terminal along the first side; and a second ground terminal disposed between the clock output terminal and the second power supply voltage input terminal along the second side.
**2**. The integrated circuit device according to claim 1, further comprising: a power supply circuit generating the power supply voltage supplied to the first circuit based on the first power supply voltage; a power supply voltage output terminal disposed along the first side and outputting the power supply voltage supplied to the first circuit; and a third ground terminal disposed between the power supply voltage output terminal and the first power supply voltage input terminal or between the power supply voltage output terminal and the clock input terminal.
**3**. The integrated circuit device according to claim 2, wherein the clock input terminal, the first ground terminal, the first power supply voltage input terminal, the third ground terminal, and the power supply voltage output terminal are arranged in this order along the first side.
**4**. The integrated circuit device according to claim 1, further comprising: an oscillation circuit oscillating a resonator to generate an oscillation signal; a buffer circuit buffering the oscillation signal and outputting the output clock signal; a first resonator terminal coupled to the resonator and the buffer circuit; and a fourth ground terminal disposed between the first resonator terminal and the clock output terminal or between the first resonator terminal and the second power supply voltage input terminal along the second side.
**5**. The integrated circuit device according to claim 4, wherein the second power supply voltage input terminal, the second ground terminal, the clock output terminal, the fourth ground terminal, and the first resonator terminal are arranged in this order along the second side.
**6**. The integrated circuit device according to claim 4, further comprising a power supply wire coupled to the second power supply voltage input terminal, wherein a shortest distance between the first resonator terminal and the power supply wire is longer than a shortest distance between the clock output terminal and the power supply wire.
**7**. The integrated circuit device according to claim 1, further comprising: a first power supply wire coupled to the first power supply voltage input terminal; a second power supply wire coupled to the second power supply voltage input terminal; and a shield wire provided between the first power supply wire and the second power supply wire.
**8**. The integrated circuit device according to claim 1, wherein the first ground terminal is coupled to an external first ground, and the second ground terminal is coupled to an external second ground

different from the first ground.
**9**. An oscillator comprising the integrated circuit device according to claim 1.
**10**. The oscillator according to claim 9, further comprising: a resonator element; and a second integrated circuit device oscillating the resonator element and outputting the reference clock signal.
**11**. The oscillator according to claim 10, further comprising a temperature control element controlling a temperature of the resonator element.